Fiscal Year:	FY 2013	**Task Last Updated:**	FY 04/25/2013
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Research Opportunities in Combustion Science		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	02/01/2010	**End Date:**	01/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:	ISS (for BASS) NOTE: End date is now 1/31/2015 per NSSC information (Ed., 3/12/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul (Universities Space Research Association) Olson, Sandra (NASA Glenn Research Center)		
Grant/Contract No.:	NNX10AD99A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A ground-based experimental and theoretical project for flame spread and extinction over solid materials is proposed with the intention of leading to a flight experiment in the space station. The research will be fundamental in nature but has practical applications to spacecraft fire safety. The objectives are (1) to determine the flame spread and growth characteristics of solids in concurrent flow in reduced gravity, (2) to determine the mechanisms of flame decay and extinction, (3) to establish a high-fidelity numerical model that can simulate both normal and reduced gravity flames, including the interpretation of the results of NASA-STD-6001A test 1. We intend to study both fabric and thick solid samples. This ground-based research is to extend the previous experimental works on solid flame spread in reduced gravity that were almost exclusively limited to very thin samples and to enhance the model capability to a three-dimensional transient formulation. The ground-based experiments will utilize both the normal-gravity and zero-gravity facilities at NASA Glenn Research center. Considerable effort will be spent to plan a space experiment that has to be carried out in the currently available facility in the International Space Station (i.e. CIR). In addition to James T'ien (PI), Dr. Paul Ferkul (NCSER) and Dr. Sandra Olson (NASA GRC) will participate as Co-Is in the project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	1. Ground-based experiment and numerical modeling Ignition of a PMMA sphere using a hot wire igniter in normal gravity followed by transition into a forced convective microgravity environment has been studied experimentally in NASA's 5.18 second drop tower and modeled using a transient numerical code. The flame is ignited at the forward stagnation region, and then the igniter is retracted. The flame is allowed to spread around the PMMA sphere before release into microgravity. The flame transitions to the new environment quickly, although the solid phase response time is longer than the available drop time. Depending on the oxygen concentration, the forced flow velocity and the degree of solid interior heating, the transition flame is either stabilized around the sphere or shrinks toward the forward stagnation point in a decaying mode. The transient numerical model reproduces the experimental sequences and provides more insight as microgravity computation can be performed in longer time durations. A direct comparison of the model and experiment has been made. The model predicts a slightly larger flame standoff distance, but the flame length and curvature is captured well. 2. Space experiment in BASS (Burning and Suppression of Solids) BASS is a rapid turnaround low budget experiment on the combustion of solid fuels currently being performed in the International Space Station. Small size samples are used in a small wind tunnel within the glove box. Among the several types of fuel samples, PMMA spheres with initial diameter of 1 and 2 cm were tested. In one of the BASS case, it took initially 25 sec for a PMMA sphere with 2-cm diameter to ignite. After some flame spread, the flame was extinguished by reducing the flow speed. When we re-ignited the sphere after some time, it took only 15 sec. Clearly there is an effect of pre-heating of the solid on ignition. In the spreading case, the initial flame is blue and weak in appearance. As the flame spread, the color becomes brighter and yellowish. This apparently stronger flame has an irregular flame shape due to fuel vapor sputtering. Both the ignition and the flame growth processes show that in-depth heating has a large effect on the flame characteristics. It should be noted that in the BASS experiment, the air is limited to ISS atmosphere which is 21±1% and one-atmosphere pressure. There is no monitoring of the internal sample temperature. In addition, the sample size is relatively small. 3. Numerical modeling of the burning of 4-cm spherical PMMA sample inside the CIR (Combustion Integrated Rig) chamber As a preparation for experiment in the CIR chamber, graduate student Makoto Endo modeled the flow pattern in the entire chamber including a recirculation tube (for flow creation) and a 4-cm diameter PMMA sphere sample. The combustion model followed Sheng-Yen Hsu's with improved parameters. Besides the flame spread and solid heat up phenomena, it showed the ability of capturing the recirculation occurring in the CIR chamber which will affect the duration of experiment. This model is going through multiple updates and currently it takes the surface regression into account. The computed results show a complex flow pattern but the flow is well behaved near the test sample. In an example, the flame grows after ignition with a 20 cm/s flow with igniter on. At 60 seconds, the igniter is turned off. The advance of pyrolysis front slows down initially but resumes its rate later. At 90 s, the flow rate is reduced in several numerical experiments. When the flow is reduced to 10 cm/s, the flame retreats initially but resumes growth with a slower rate later. If the flow is reduced to 2 cm/s, the flame shrinks and becomes extinct. At the intermediate 7 cm/s, the flame front is stationary thus yielding a dividing limit between flame growth and extinction. This is of course a function of internal heating. This simulation demonstrated the capability that with computational fluid mechanics and computational combustion, the flow pattern in the entire test chamber can be included in the study of the combustion behavior.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Abstracts for Journals and Proceedings	T'ien JS, Endo M. "Material flammability: a combustion science prospective." 9th Asia-Oceania Symposium on Fire Science and Technology, Hefei, China, Oct. 17-21, 2012. 9th Asia-Oceania Symposium on Fire Science and Technology, Hefei, China, Oct. 17-21, 2012. http://202.38.64.11/~aosfst2012/9thAOSFST_Paper_Abstracts.pdf , Oct-2012
Articles in Peer-reviewed Journals	Hsu SY, T'ien JS. "Flame spread over solids in buoyant and forced concurrent flows: Model computations and comparison with experiments." Proceedings of the Combustion Institute. 2011;33(2):2433-40. http://dx.doi.org/10.1016/j.proci.2010.05.093 , Dec-2011
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Articles in Peer-reviewed Journals	Kumar A, T'ien JS. "Numerical modeling of limiting oxygen index apparatus for film type fuels." International Journal of Spray and Combustion Dynamics. 2012 Dec;4(4):299-322. http://dx.doi.org/10.1260/1756-8277.4.4.299 , Dec-2012

Articles in Peer-reviewed Journals	T’ien JS, Endo M. "Material flammability: a combustion science prospective." Procedia Engineering. 2013;62:120-9. http://dx.doi.org/10.1016/j.proeng.2013.08.049 From: 9th Asia-Oceania Symposium on Fire Science and Technology, Hefei, China, Oct. 17-21, 2012. , Jan-2013
Papers from Meeting Proceedings	Olson S, Johnston M, Hsu S, T’ien JS, Ferkul PV. "Experiments and Modeling of Normal Gravity Ignition and Transition to Microgravity for a PMMA Sphere in a Forced Convective Flow." 7th US National Technical Meeting of the Combustion Institute, Atlanta, GA, March 20-23, 2011. In: 7th US National Technical Meeting of the Combustion Institute, Georgia Institute of Technology, Atlanta, GA, March 20-23, 2011. Pittsburgh, Pa. : The Combustion Institute, 2011. p. 1729-1735. Printed by Curran Associates, Inc. ISBN: 978-1-61782-780-8. , Mar-2011